

Problem Statement

The problem of smallholder farmers accessing timely, localized, and linguistically appropriate climate-agricultural guidance has been a long-standing issue across sub-Saharan Africa. In Nigeria specifically, where agriculture employs over 70% of the rural population and contributes significantly to national GDP, farmers increasingly face unpredictable rainfall patterns, shifting planting seasons, and novel pest outbreaks driven by climate change. Current solutions involve federal and state agricultural extension services and general-purpose mobile applications, which pose significant challenges including inadequate coverage of rural communities, reliance on English-only interfaces, poor connectivity assumptions, and the absence of hyperlocal climate context relevant to specific agroecological zones.

This research aims to develop a small language model (SLM) fine-tuned on Nigerian agricultural data, regional climate records, and indigenous farming knowledge, and deployable in major local languages including Yoruba, Hausa, and Igbo. The system will target low-bandwidth environments to remain accessible to rural farmers.

The successful development of this approach will improve farmer decision-making around planting schedules, pest and disease management, and climate adaptation strategies, directly strengthening food security and livelihood resilience for Nigeria's most climate-vulnerable communities, with lessons transferable across the African continent.